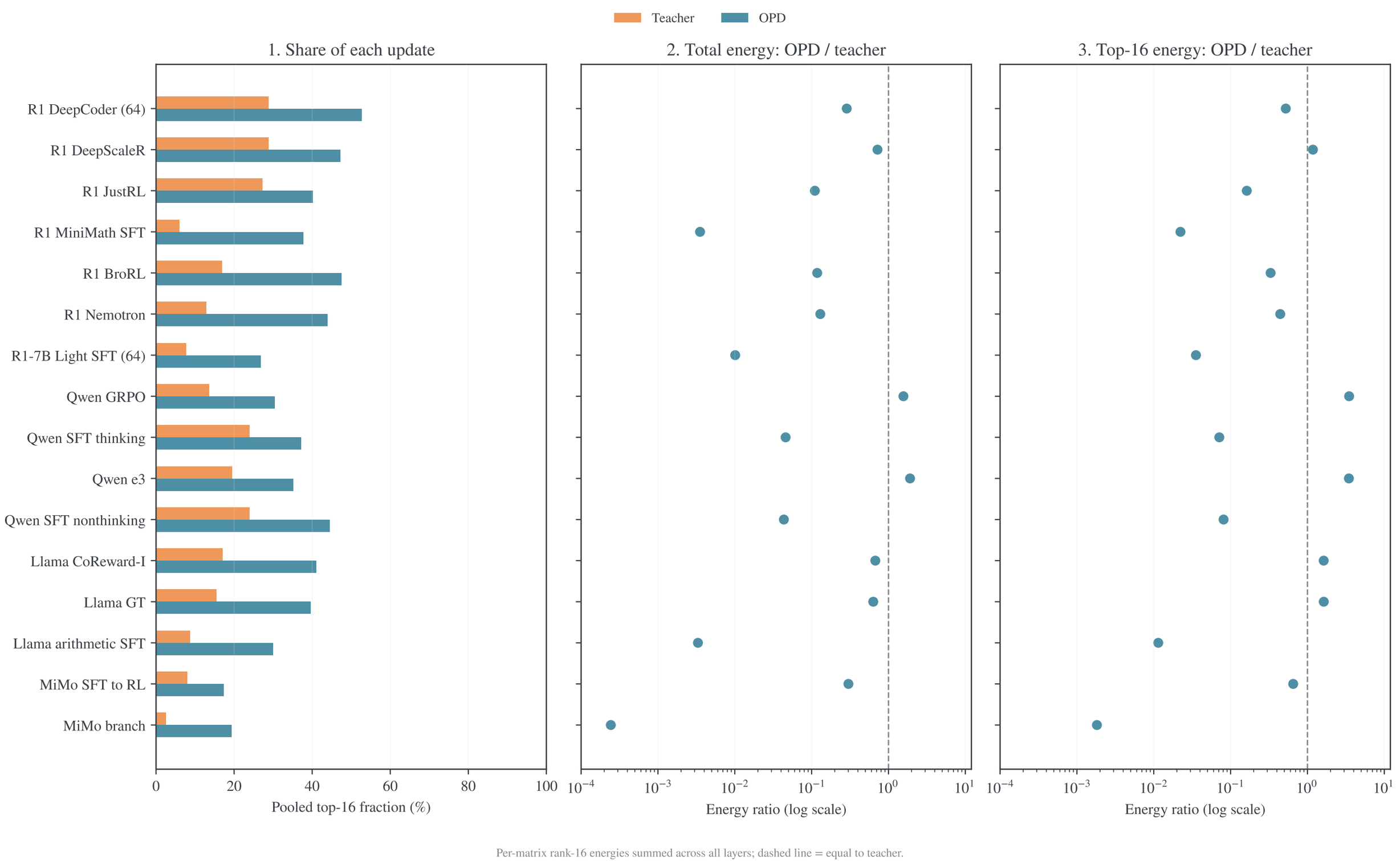
Teacher
OPD
1. Share of each update
2. Total energy: OPD / teacher
3. Top-16 energy: OPD / teacher
R1 DeepCoder (64)
R1 DeepScaleR
R1 JustRL
R1 MiniMath SFT
R1 BroRL
R1 Nemotron
R1-7B Light SFT (64)
Qwen GRPO
Qwen SFT thinking
Qwen e3
Qwen SFT nonthinking
Llama CoReward-I
Llama GT
Llama arithmetic SFT
MiMo SFT to RL
MiMo branch
0
20
40
60
80
100
Pooled top-16 fraction (%)
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
Energy ratio (log scale)
Energy ratio (log scale)
Per-matrix rank-16 energies summed across all layers; dashed line = equal to teacher.